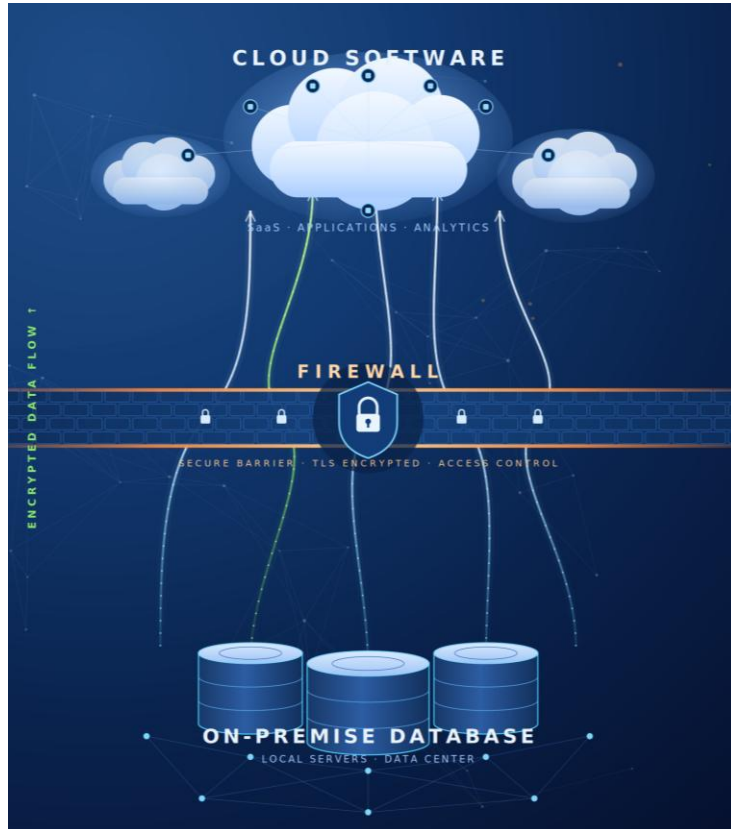


Selective Data Replication from S/4 HANA into HANA Cloud



Agenda

- 1. The Project**
to substitute an entire SAP module with a custom tailor-made SAP BTP solution
- 2. The Connection**
between HANA Cloud in BTP and the HANA DB behind S/4
- 3. The Replication**
of selected data and fields from S/4 to update what changed in HANA Cloud
- 4. The Question**
can this be considered a long-term best practice of a Side-by-Side Extension to S/4?

By Julian Mayer – BTP & HANA Cloud Solution Architect from cbs in co-operation with Leonhard Weiss

CLOUD SOFTWARE

SaaS APPLICATIONS · ANALYTICS

FIREWALL

SECURE GATEWAY · TLS ENCRYPTED · ACCESS

ON-PREMISE DATABASE

LOCAL SERVERS · DATA CENTER

The Project

to substitute an entire SAP module
with a custom tailor-made SAP BTP solution



Who we are

Leonhard Weiss

A family-owned construction company founded in 1900 and today one of Germany's largest. Three divisions — road & network construction, rail infrastructure, and engineering & turnkey building — deliver everything from single trades to complex turnkey works.



LEONHARD WEISS
BAUUNTERNEHMUNG



1900

Founded — originally rail track

~7,700

Employees (2024)

€2.4 bn

Construction output (2024)

Göppingen

HQ · Baden-Württemberg

The implementation partner

cbs Corporate Business Solutions

A global SAP consultancy and part of the Materna Group, specializing in end-to-end business processes on SAP and cloud technology. cbs partnered with Leonhard Weiss to design and build the custom BTP solution and the HANA replication shown here.



1995

Founded · Heidelberg

~1,700

Employees worldwide

40+

Locations globally

Materna

Member of Materna Group

The project

Replacing SAP ETM with a custom tailor-made BTP solution

SAP ETM — SAP's Equipment & Tools Management for the construction world — reached end of maintenance at the end of 2025. Rather than a like-for-like product swap, Leonhard Weiss built a full-fledged custom solution on SAP BTP and put the power of SAP HANA to work.



SAP ETM

End of maintenance · 2025



Full-fledged BTP solution

Rebuilt as our own application on SAP BTP — our processes, our data model, our control.



Replicated Master Data

A selective subset of master data is replicated from S/4HANA into our own HANA Cloud.



Managed Moves

Smoothly managing the Articles how they move between Holders (warehouses, storage locations and construction sites)

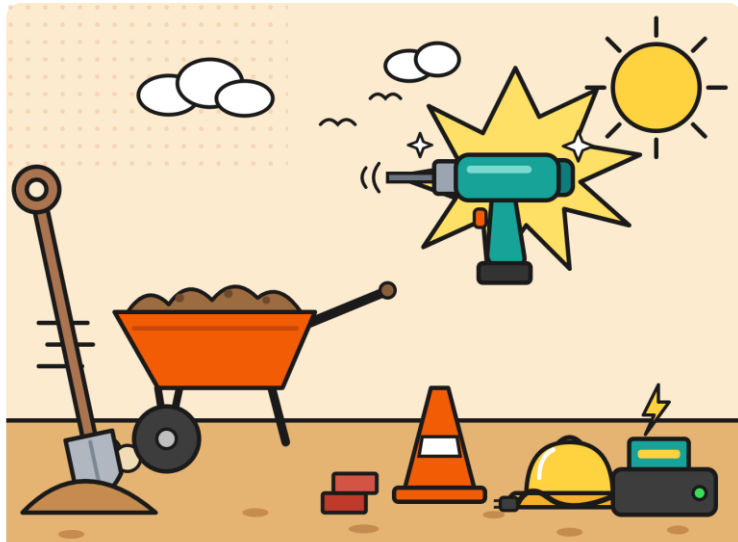


Tracked and Charged

Tracking all movements and calculating based on customizable accounting types who gets charged how much

S/4HANA Materials become Articles (Items used on Construction Sites)

Our HANA Cloud application manages **Articles** — the items consumed and tracked on a construction site. A fair portion of them are grounded on a **Material Master (MARA)** from S/4. Others in a similar way from the **Equipment Master (EQUI)**. All must be pulled into our solution to become **Articles**.



Field mapping · MARA / MAKT → Article

MATNR	→ NAME <i>leading zeros trimmed</i>
MAKTX	→ DESCRIPTION <i>German text, SPRAS = 'D'</i>
MEINS	→ UNITOFMEASURE_CODE <i>base unit of measure</i>
MATKL	→ FOLDER assignment <i>material group → hierarchy</i>

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The Connection

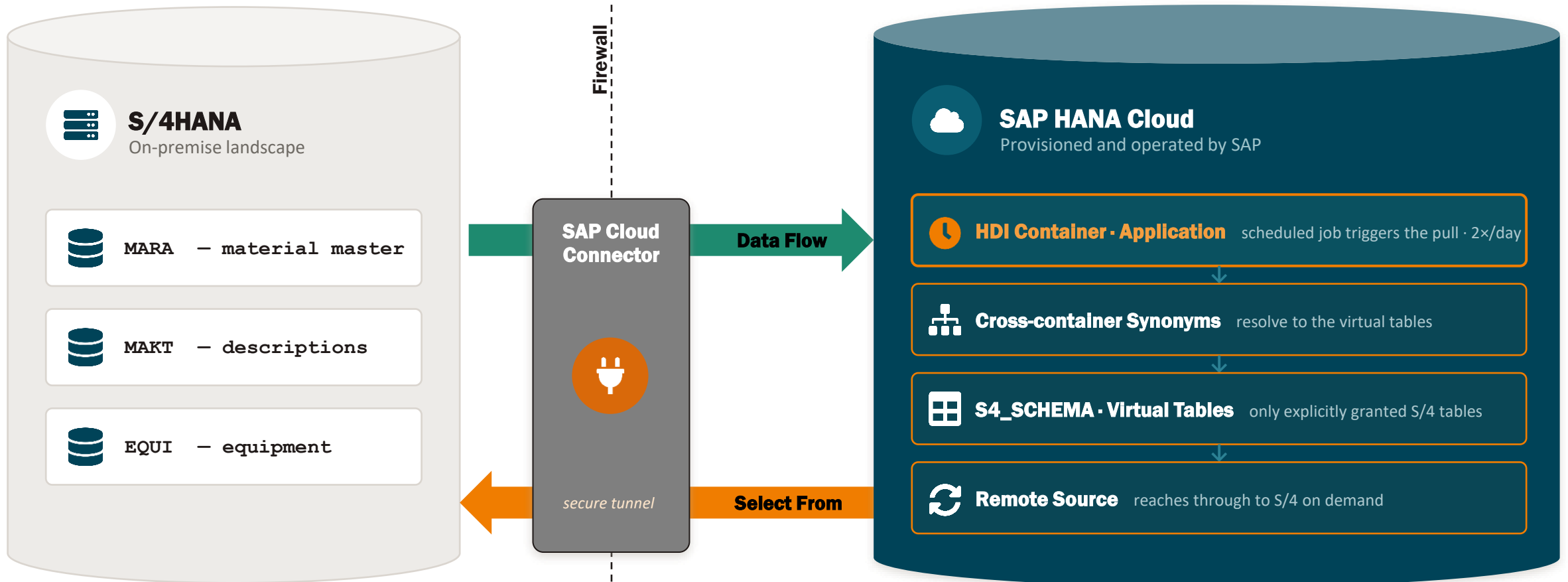
between HANA Cloud in BTP
and the HANA DB behind S/4

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Smart Data Access across the Cloud Connector



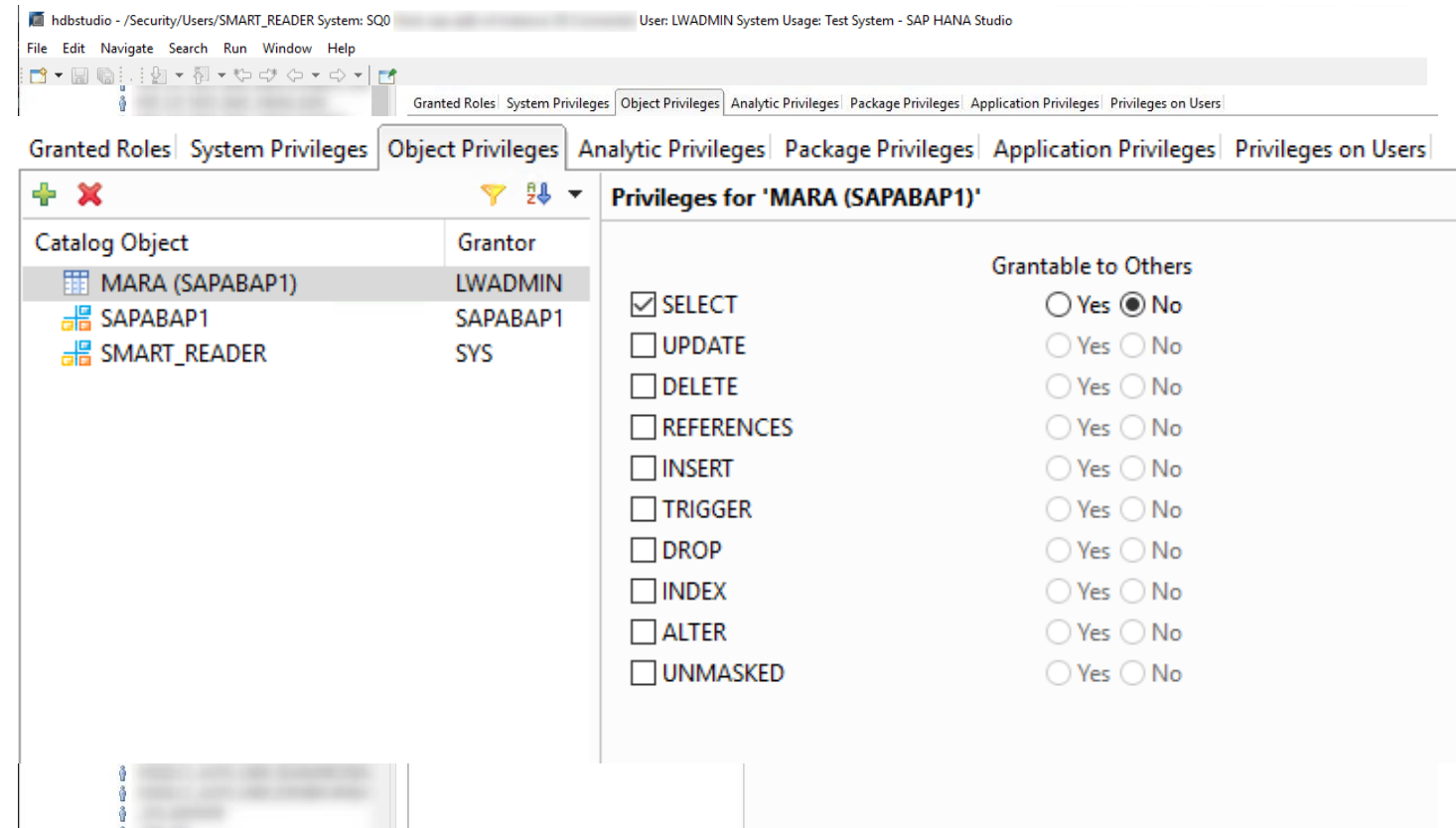
Pull, not push. The HDI scheduled job issues the SELECT through the synonym → virtual-table chain; S/4 returns only the requested rows, on demand. The on-prem system is never the initiator and is not required to run, provide or install anything.

Securely selecting from HANA DB behind S/4

1 Create a Technical DB User in HANA Studio

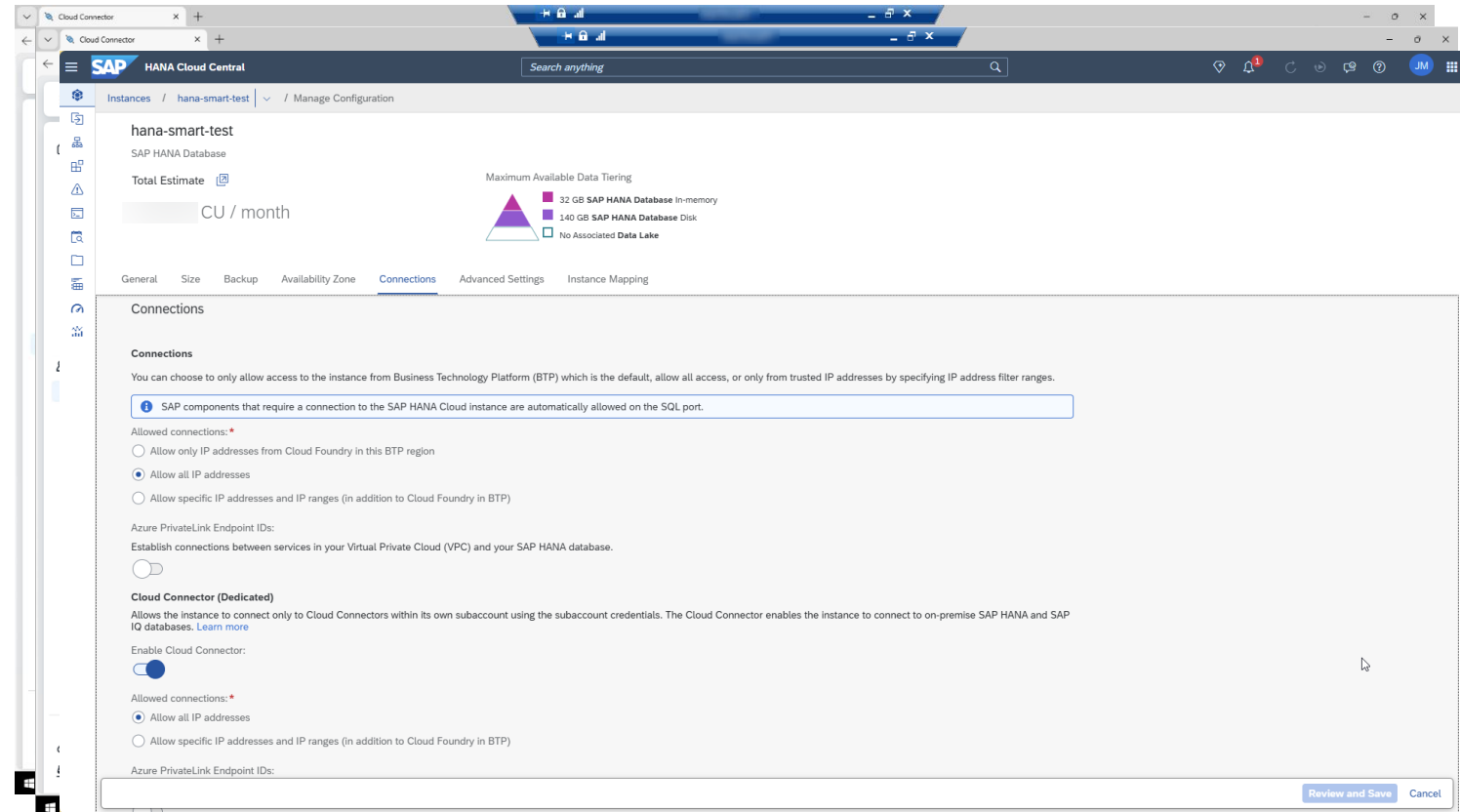
2 Grant SELECT Privilege to entire S/4 Schema

3 Or grant SELECT for selected Tables



Few steps, and the tunnel is live

- 1 Install CC on-prem · link to BTP subaccount
- 2 Expose the on-prem HANA DB as a virtual host
- 3 Enable Cloud Connector on the HANA Cloud instance



Locked down by design — nothing else is reachable



Remote Source created manually

A single, deliberate connection to the S/4 system using a technical user with SELECT from S/4 Schema privileges. Either for all tables or a dedicated set of allowed Tables



Virtual tables in a dedicated S4_SCHEMA

The SAP Basis admin provisions exactly the tables we need — MARA, MAKT, EQUI — and nothing more.



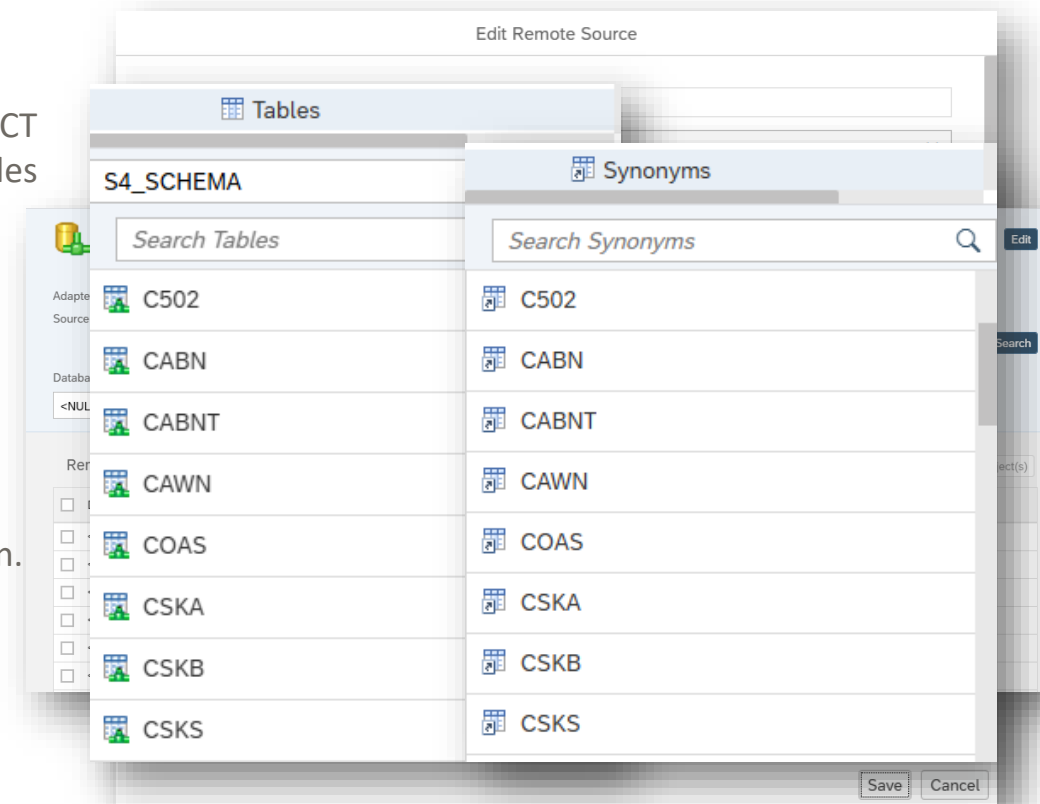
Explicit creates only (& grants on HANA DB behind S/4)

If a virtual table was not created, it simply does not exist for us. No path to reach unrelated S/4 data. Limiting the granted tables may feel safer but also slows down.



Cross-container synonyms

Each HDI container sees the virtual tables through synonyms, keeping app code clean and portable.



Granted → virtual tables

MARA · MAKT · EQUI · reachable



Everything else in S/4

not granted · not exposed · no access

Our **one stop shop** for reading S/4 data

Project is growing, Scope is expanding, dependencies to S/4 are surfacing — just add a virtual table and carry on

WITHOUT SDA

"We need the plant data from S/4 too." -> raise API ticket -> architecture review -> wait weeks -> hope nothing changed

VS

WITH SDA

"We need the plant data from S/4 too." -> grant one virtual table -> write the SELECT -> done



Zero lead time

When a new S/4 table is needed, no API team, no middleware change, no ticket queue — a virtual table grant is the only gate.



Consistent, versioned access

All data access lives in one schema, owned by one team. No fragmented endpoints, no sync drift between sources.



Keeps pace with a growing scope

In a live project where dependencies surface weekly, the read layer never becomes the bottleneck that slows discovery.

The connection was there. We just added a table.

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The Replication

of selected data and fields from S/4
to update what changed in HANA Cloud

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One pattern drives every replication job

Each procedure follows the same four set-based steps — read, reshape, diff, write — all pushed down into HANA.

1



Extract & Filter

Select the relevant rows straight from the S/4 tables and apply business filters.

2



Transform & Enrich

Reshape into the target model and enrich with data that already exists in HANA.

3



Detect Changes

Use EXCEPT to compute the exact set of new and changed rows — and only those.

4



UPSERT to Target

Write the delta into the target table in a single, idempotent statement.

Why this scales

A simple recipe that runs the whole estate



Pushdown by default

All logic is set-based and runs in HANA, right next to the data.



One Interface - many Objects

The same four steps power dozens of replication procedures.



Cheap, exact deltas

EXCEPT finds changes without row-by-row comparison.



Idempotent & re-runnable

UPSERT and MERGE converge to source state, with soft deletes.



Governed access

A dedicated S4_SCHEMA and synonyms expose only what's granted.

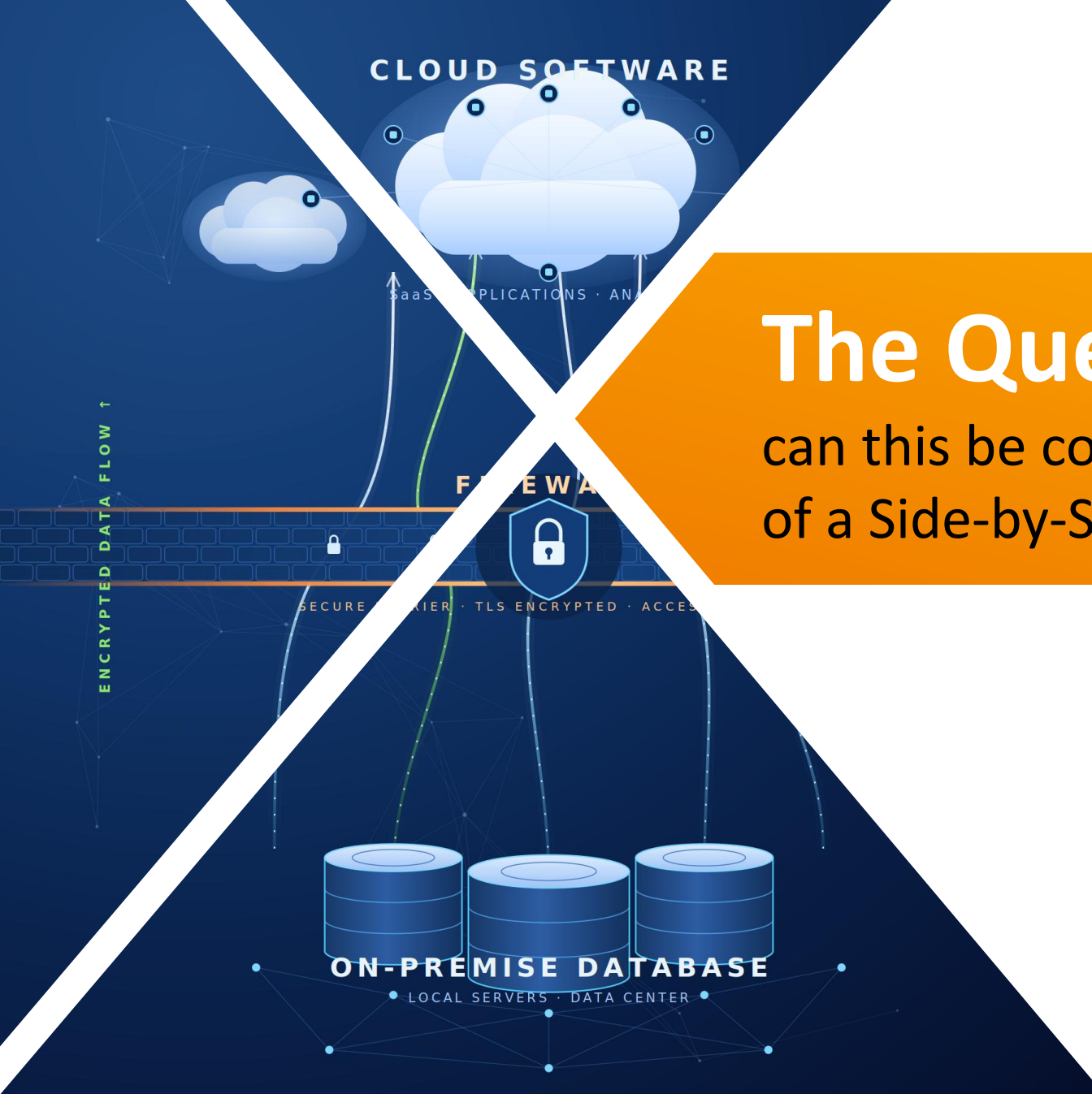


No external ETL burden

HANA scheduled jobs run it twice a day — that's the whole pipeline.

Thank you — questions welcome.





The Question

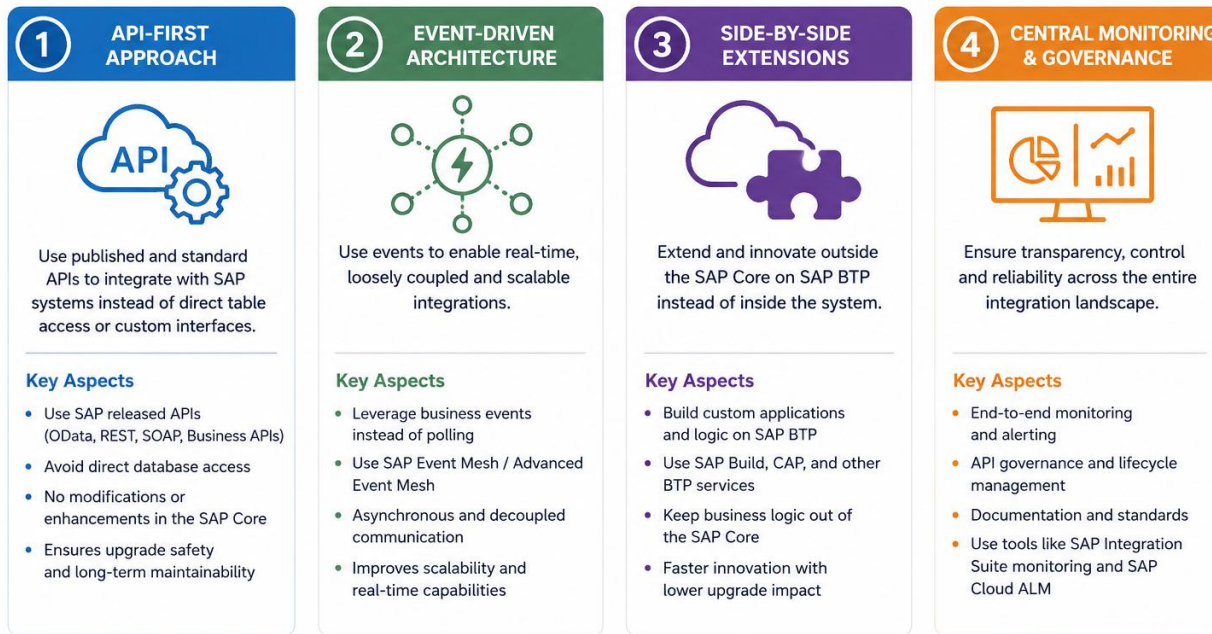
can this be considered a long-term best practice of a Side-by-Side Extension to S/4?



Is Side-by-Side with select from DB tables still Clean Core compliant?

The 4 Core Principles of Clean Integration

Design integrations that keep the SAP Core clean, upgradeable and future-ready



Clean Integration enables a clean SAP Core, reduces technical debt and accelerates innovation.

Integrate via standard APIs, leverage events, build side-by-side and govern end-to-end.

1

It can't break anything inside of S/4 or even block an Upgrade

2

If a database field would change, only one replication peace would fail on cloud side

3

Extreme low resource consumption for mass date transfer on both sides

4

Any API alternative posts far higher cost of ownership and implementation



Yes, we see it as Clean Core compliant!
How do you (SAP) see it?

Contact



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Do you consider a Make instead of a Buy?

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